

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458207.6362 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.053
Transit depth (Rp/Rs)^2	3.91 +/- 2.11 [%]
Orbital Inclination (inc)	81.9 +/- 2.33
Airmass coefficient 1 (a1)	1.005 +/- 0.024
Airmass coefficient 2 (a2)	0.001 +/- 0.063
Scatter in the residuals of the lightcurve fit is	2.39 %
Best Comparison Star	#2 - [336, 45]
Optimal Aperture	2.85
Optimal Annulus	7.01
Transit Duration (day)	0.0327 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.198 ± 0.053 vs 0.1594 (deviation 0.73σ)
Duration fit vs published	0.0327 d vs 0.0512 d (deviation 3.5σ)
Mid-time O-C	-3.8 min
Depth SNR	3.7
Residual scatter	2.39 %

Light curve

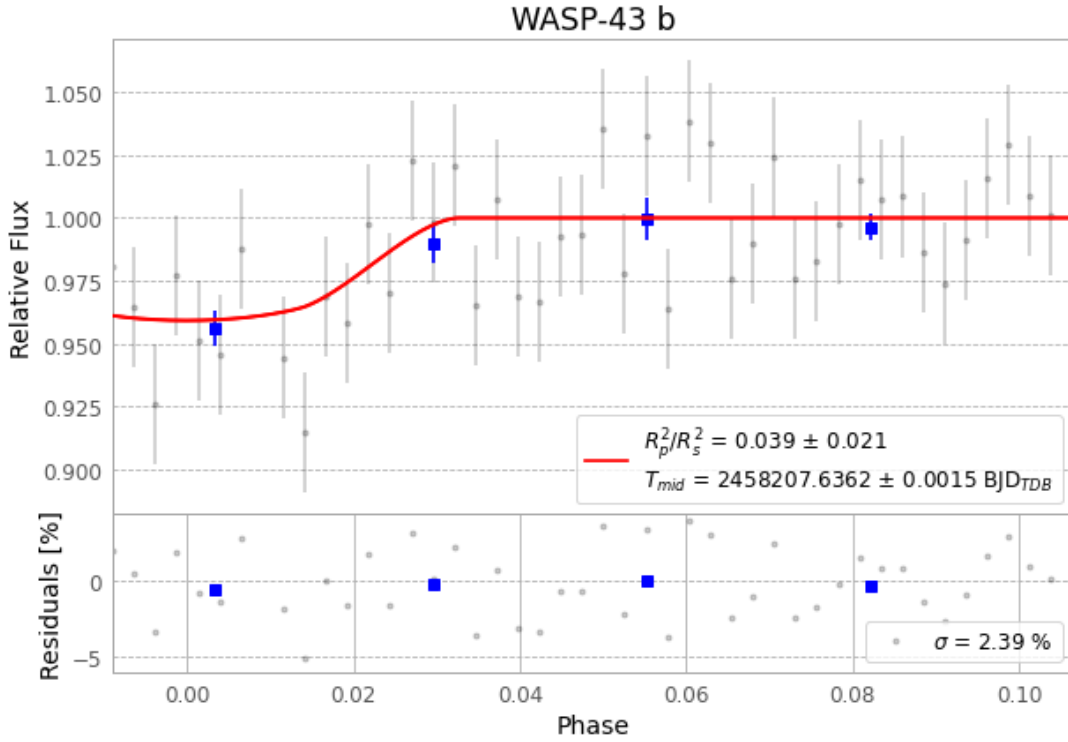


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-29.png (SHA-256 cb4afdc4789669bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [456, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8541d9a3dfd0486d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

46 FITS frames (30 MB), WASP-43180330025712.FITS → WASP-43180330051212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180330.log (5da707ea202b...), FinalLightCurve_WASP-43 b_2018-03-29.png (cb4afdc47896...), FinalLightCurve_WASP-43 b_2018-03-29.csv (f9d4b5ead2ff...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 05:05Z.